Redundant Connection (medium)

Problem Statement

Given an undirected graph containing `1` to `n` nodes. The graph is represented by a 2D array of `edges`, where **edges[i]** = [a_i, b_i], represents an edge between a_i, and b_i.

Identify one edge that, if removed, will turn the graph into a tree.

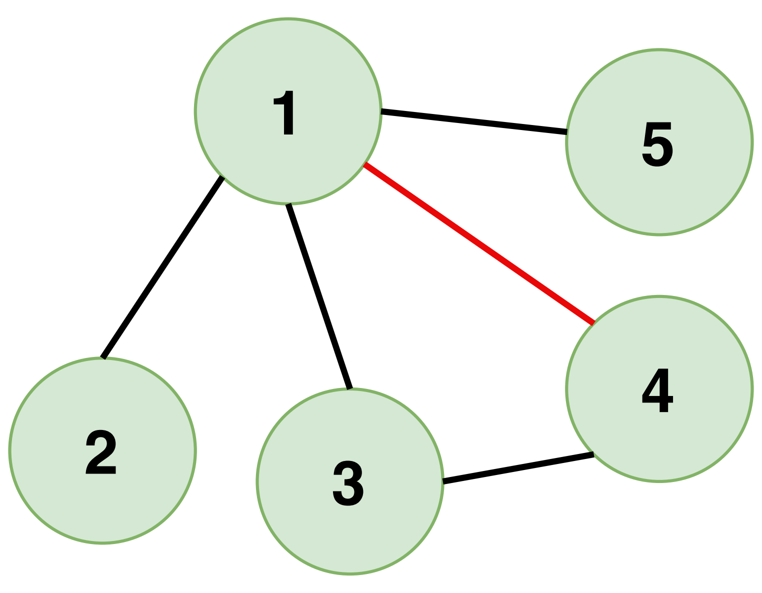
A tree is a graph that is connected and has no cycles.

Assume that the graph is always reducible to a tree by removing just one edge.

If there are multiple answers, return the edge that occurs last in the input.

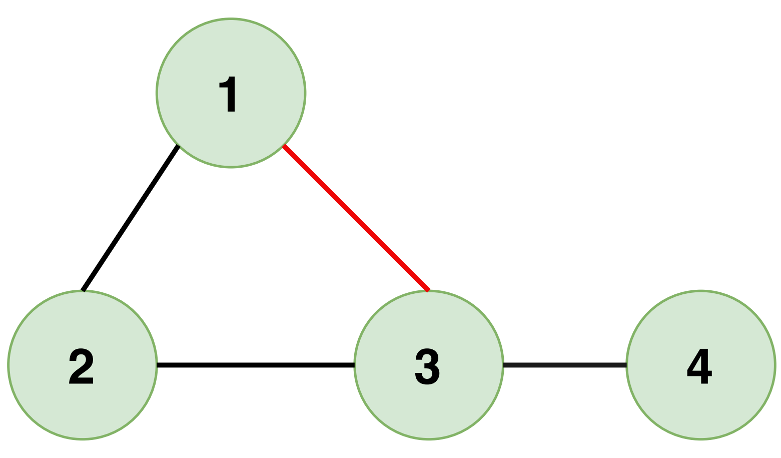
Examples

1. Example 1:
    - **Input**: `[[1,2], [1,3], [3,4], [1,4], [1,5]]`



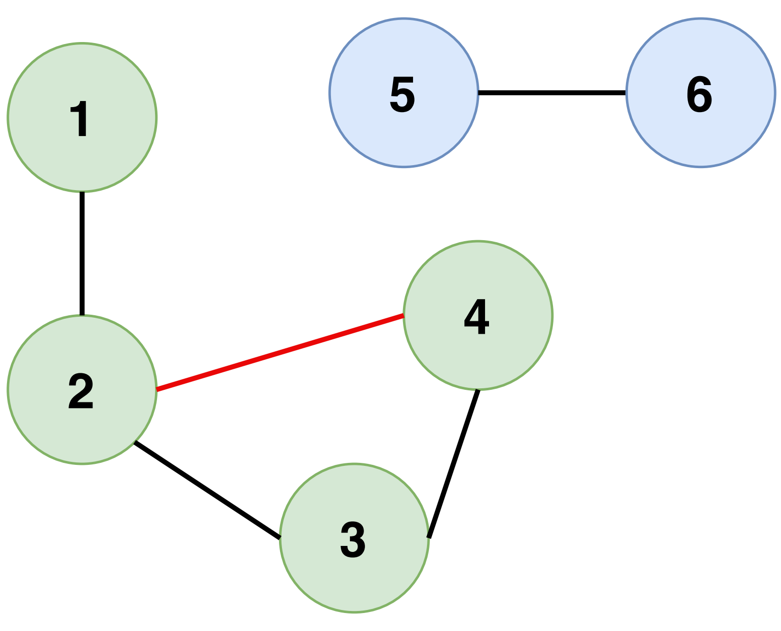
- **Expected Output**: `[1,4]`
    - **Justification**: The edge `[1,4]` is redundant because removing it will eliminate the cycle `1-3-4-1` while keeping the graph connected.

2. Example 2:
    - **Input**: `[[1,2], [2,3], [3,1], [3,4]]`



- **Expected Output**: `[3,1]`
- **Justification**: The edge `[3,1]` creates a cycle `1-2-3-1`. Removing it leaves a tree with no cycles.

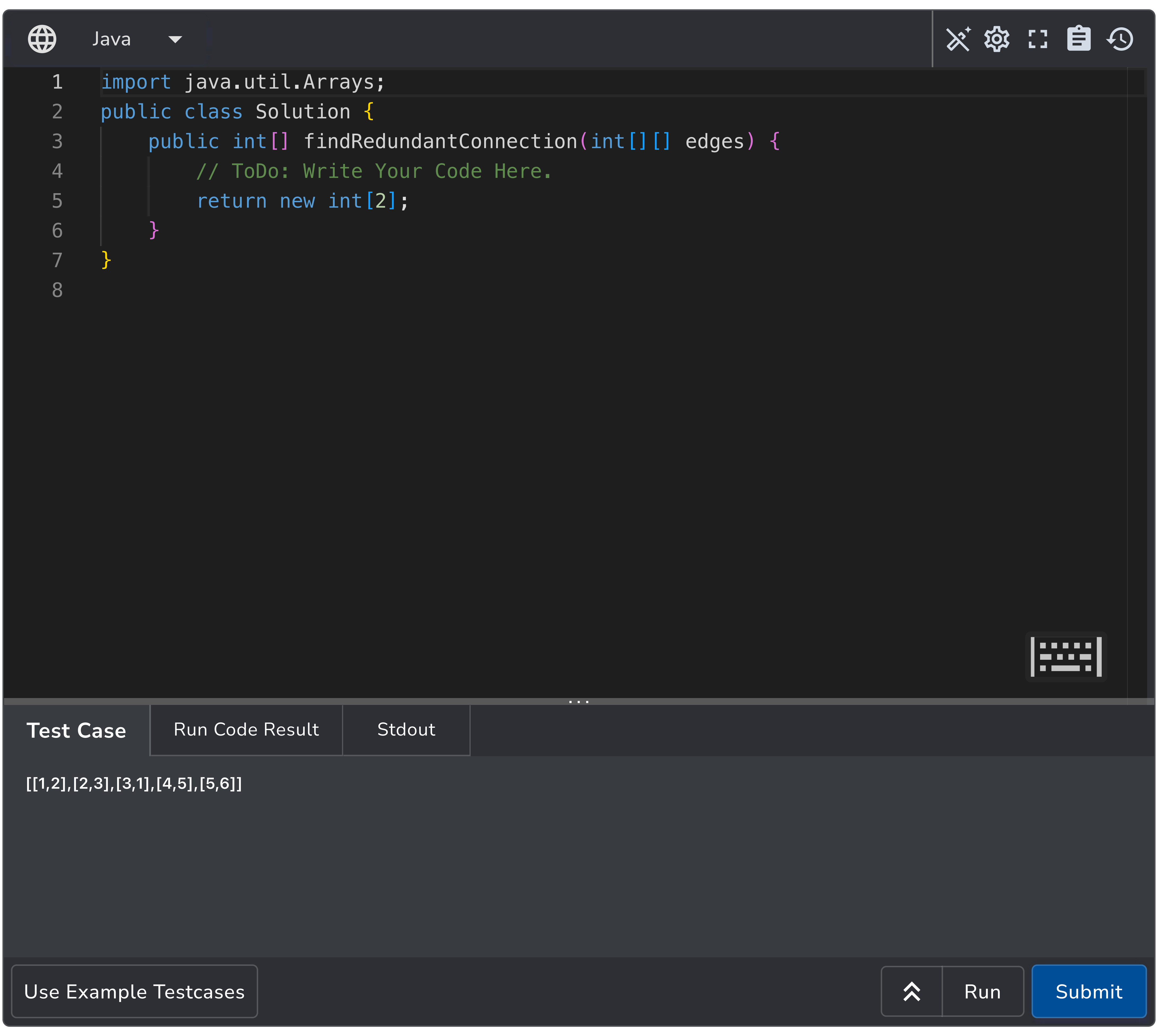
3. Example 3:
    - **Input**: `[[1,2], [2,3], [3,4], [4,2], [5,6]]`



- **Expected Output**: `[4,2]`
- **Justification**: The edge `[4,2]` adds a cycle `2-3-4-2` in one part of the graph. Removing this edge breaks the cycle, and the remaining graph is a tree.

Try it yourself

Try solving this question here:



```java
import java.util.Arrays;
public class Solution {
    public int[] findRedundantConnection(int[][] edges) {
        // ToDo: Write Your Code Here.
        return new int[2];
    }
}
```

[[1,2],[2,3],[3,1],[4,5],[5,6]]

← Previous
Introduction To Union Find Pattern

Next →
Solution: Redundant Connection

☐ Mark as Completed